B. Tech. (3rd Sem)

(Common for CSE, CSE with Specialization in Data Science, CSE with specialization in Cloud Technology & Information Security, CSE with specialization in Big Data and Analytics, CSE with specialization in Full Stack Development)

BCSE-502L (Software Engineering Lab)

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0	0	2	Continuous evaluation	60
			End semester exam	40
			Total marks	100
			Credits	1.0

Course Objectives*:*

1. Identify complex engineering problems by applying principles of software engineering, and testing.
2. To gain software testing experience by applying software testing knowledge and methods to practice-oriented software testing projects.
3. To provide an idea of using various testing techniques in the software industry.
4. To analyze the processes and products to be tested and use engineering judgment to draw conclusions.

List of Practical

1. Create Software Requirement Specification (SRS) of Bank Application Management.
2. Create Use Case, Class and object diagram of Bank Application Management.
3. Implement Activity, Sequence Diagram for Bank Application System.
4. Create RTM (Requirement Traceability Matrix) for the given scenario.
5. Create ETVX model for the given scenario.
6. Create test cases for ATM transactions using any White Box technique.
7. Create test cases for ATM transactions using any Black Box technique.
8. **Tool Installation:** 1) JIRA 2) MANTIS 3) Bugzilla
   Task : Analysis Report : As you have gone through all the three tools you need to now compare and contrast the similarities and the differences between these tools and find which tool will be relevant for which kind of projects .You need to create the analysis report based on your research and tool knowledge you have gained.
9. **Unit Testing Tools - Junit :** Create a program of sum of numbers. Create test cases for the same using junit. Get the result with the help of test runner.
10. **Defect Management Tools – Mantis /Bugzilla /Zira :** Do the following using MantisBT / Bugzilla
    Create a Project named Prj<Toolname><YourName>
    1. Create four users - Administrator, Tester 1, Tester 2 & Developer1 and add to the project created.
    2. Create a custom field and link it to the project created.
    3. Create 4 bugs (of different severity & priority) using the tool in the project created.
    4. Assign these bugs to the administrator and/or developer.
11. **TESTING PROJECT : ATM Simulator Project**
    Project Instructions: Download "ATM Simulation Project.zip" and load on your Eclipse.
    1. You will work as a Testing team in an organization.
    2. Pick a name for your Team
    3. Each one of you in the team would be playing one of the 3 roles in a Testing Project **- Test Manager, Test Analyst and Tester.**
    4. Find a set of **Test Plan Templates**. Use them to create your deliverables.
    5. You will also be required to give a presentation at the end of your project as a team.

    Overview: The Project has a four-phase exercise work that consists of the followingassignments:
    1. Phase 1: Test case design (individual assignment)
    2. Phase 2: Test execution and reporting (individual assignment)
    3. Phase 3: Test planning (group assignment)
    4. Phase 4: Presentation (group assignment)

    Exercise phases: Exercise phases are briefly described below:

    Phase 1: Test case design (individual assignment): In the first and second phase you will do system level functional testing for the ATM Simulator Project. In the first phase the test cases are designed using several test design techniques that you have learned. You will create 10 Test cases and log them in the Testcase template. You have to create and run any 5 test cases using JUnit.

    Phase 2: Test execution and Reporting (individual assignment): In the second phase the tests are executed and test results reported. Use the template for Defect Log and Test Summary Report to submit these.

    Phase 3: Test planning (group assignment): In this phase, the system testing of the ATM Simulator project is planned in groups. At this point you should have adequate skills and knowledge of testing techniques and tools as well as experience of using the software under test in order to do the planning work. In this phase you will use this knowledge to write a Test plan document. In this phase, we have a fictional case description that outlines the context where the testing project is conducted and gives background information to assist you in test planning. You will also create RACI matrix and Requirements Traceability Matrix in this phase.

Phase 4: Presentation (group assignment): In phase 4, student groups perform a formal presentation of a selected document and how the group went about performing the exercise. In the presentation session, only 15 minutes is reserved per team.

Tools Required: In Exercise Phases 1-2, the program to be tested is the ATM Simulator. ATM Simulator is an applet written in Java. JUnit should be used for creating any 5 test cases in the exercise phase 1.MantisBT should be used for reporting defects in the exercise phase 2.Test plans and reports should be written using standard word processing and spreadsheet software (e.g. Word and Excel). Accepted document delivery formats are Word (doc), pdf, and Excel formats.

12. **Automation Testing:** Install Test Automation Tool - TestCompletePick a Desktop application (of your choice) and perform the following Automation tutorial.
13. **TEST AUTOMATION PROJECT :** Web Automation Test Project

Test Scenario

For this project, you need to use TestComplete automation tool to test the dummy web app accessible at https://www.phptravels.net. This website is a travel portal that allows customers to book hotels, flights, tours etc.

Note: The demo *usernames and passwords* are below:

	Url	Demo Username	Demo Password
Homepage Front-End	https://www.phptravels.net	user@phptravels.com	demouser
Administrator Back-End	https://www.phptravels.net/admin	admin@phptravels.com	demoadmin
Supplier back-End	https://www.phptravels.net/supplier	supplier@phptravels.com	demosupplier

In this project, you have to create test cases to test the following flows:

1. **Scenario 1**
   - Login as a Customer
   - Book a hotel
   - Confirm booking
2. **Scenario 2**
   - Login as an Administrator
   - Go to "Manage Offers"
   - Add an Offer
   - Logout as Administrator
3. **Scenario 3**
   - Login as an Supplier
   - Choose a workflow of your choice and create a test
4. **Scenario 4**
   - Pick any one scenario from above and create Cross Browser Tests for that scenario

Course Outcomes*:* After the completion of this course, the students will be able to:

1. Formulate, solve and test complex engineering problems by applying principles of software engineering, and testing.
2. Conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
3. Demonstrate an ability to use the tools necessary for engineering practice.
4. Apply different types of testing for real life